Tim Perr

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Profile

Graduate student passionate about scientific computing and exploring the intersection of computing, astronomy, biology, and physics. I enjoy tackling complex problems, working with data, and diving into systems-level programming, operating systems, and AI related projects. I'm excited about contributing to projects where science, technology, and creativity come together to make an impact.

Education

Master of Science in Computer Science
Michigan Technological University Aug 2024–Dec 2025
Master of Science in Data Science
Michigan Technological University Aug 2024–Dec 2025
Bachelor of Science in Computer Science
Michigan Technological University Aug 2021–Apr 2024

Skills

Programming Languages: Python, C/C++, Java, Swift, R, Matlab, Bash, MIPS Assembly, Makefile, LaTeX
Web Development: React.js, Node.js, HTML, CSS, JavaScript, Flask, PHP
Data and Machine Learning: SQL, Data Analysis, Machine Learning, Python libraries (e.g., NumPy, Pandas, Matplotlib), MATLAB Toolkits
Systems and Theoretical Knowledge: Concurrent Computing, Quantum Computing, Discrete Mathematics, Programming Language Design, Computing Theory
Mathematics and Science: Linear Algebra, Calculus, Statistics, General Physics and Astronomy
Artificial Intelligence: Various AI methods, AI in Healthcare, Computer Vision

Work and Research Experience

Computer Science Research Assistant
Michigan Technological University Feb 2023 – Present
- Contributed to the development of a website for Illuminated Devices, supported by an NSF-funded project #BCS-2122034.
- Presented research on the theoretical aspects of the project at ASEE 2024; view the paper https://peer.asee.org/46774.

Spinel Lab Member
Michigan Technological University Nov 2024 – Present
- Focused on Computer Science Education, specifically in reframing perceptions of computing and developing supplemental learning materials.
- Responsible for data analysis and rewriting homework questions to enhance learning materials.

Computer Science Learning Center Coach
Michigan Technological University Sep 2022 – Apr 2024
- Guided students through coursework across various levels, from introductory to graduate computing courses.

Computer Science Lab Assistant
Michigan Technological University Sep 2022 – Dec 2022
- Assisted in teaching introductory computer science labs (CS1111, CS1131, CS1122).

Instructor
iD Tech Camps May 2022 – Aug 2022
- Delivered curriculum to classes of up to 12 students, ensuring engagement and understanding of programming concepts.
- Monitored students' progress toward learning goals and facilitated non-instructional activities.

Bank Teller
First National Bank of LaGrange Oct 2018 – Aug 2021
- Processed cash transactions, balanced drawers and vaults, issued cashier's checks, and delivered exceptional customer service.

Projects

Entropy Emulator
Objective: Develop a simple emulator for a computer
- Programmed in C++ with a non-complex Makefile.
- Reads a file of computer operations (clock tick, memory load, CPU load) and executes commands with assembly-like operations.
- Includes operations such as Load/Save Word and Branches.
- https://github.com/tperr/Entropy-Emulator

Fracture Detector
Objective: Use a neural network to detect fractures
- Final project for SAT5114: AI in Healthcare.
- Implemented in Python using U-Net architecture.
- Trained on a dataset provided by figshare.com.
- https://github.com/tperr/Fracture-Detector

Certificates

Essentials of Free and Open Source Software
Canvas Credentials (Badgr) 08/2024
https://api.badgr.io/public/assertions/EXNMMFfHTc27lXLTj2oS1w
Deep Learning Onramp
Mathworks 11/2024
https://matlabacademy.mathworks.com/progress/share/certificate.html?id=ee0531f0-bb8a-408c-8604-22fd76a5783d&